Project Title

E-Commerce Product Entity-Resolution & Matching Platform

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

Submitted by
Hrushika Habeeb (2520090054)
Chakrika Panakala (2520090042)

Team Number: 20

Section: 12

Under the guidance of

Ms. K. Chandusha

Assistant Professor, CS-IT

Hyderabad-500090,

Telangana, India.



K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING / COMPUTER SCIENCE AND INFORMATION TECHNOLOGY/ARTIFICIAL INTELLIGENCE & DATA SCIENCE



DECLARATION

The Mini Project Report entitled "E-Commerce Product Entity-Resolution & Matching Platform" is a record of Bonafide work of Hrushika Habeeb – 2520090054 and Chakrika Panakala – 2520090042 submitted in partial fulfillment for the award of B. Tech in Computer Science and Information Technology to the K L (Deemed to be) University. The results embodied in this report have not been copied from any other department/University/Institute.

Hrushika Habeeb – 2520090054

Chakrika Panakala – 2520090042

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING / COMPUTER SCIENCE AND INFORMATION TECHNOLOGY



CERTIFICATE

This is to certify that the mini project based report entitled "E-Commerce Product Entity-Resolution & Matching Platform" is a bonafide work done and submitted by Hrushika Habeeb – 2520090054 and Chakrika Panakala – 2520090042 in partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in Computer Science and Information Technology, K L (Deemed to be) University, during the academic year 2026-2027.

Signature of the Supervisor

Signature of the HOD **Signature of the External Examiner**

ACKNOWLEDGEMENT

The success in this project would not have been possible but for the timely help and guidance rendered by many people. Our wish to express my sincere thanks to all those who has assisted us in one way or the other for the completion of my project.

Our greatest appreciation to our guide, Ms. K. Chandusha, Assistant Professor, CS-IT, for her tremendous support, encouragement and guidance for this project.

We express our gratitude to **Dr. P.Venkateshwar Rao (CSE)/Dr. Khaja Shareef (CSIT)/ Dr. Sreenath Itikela (AI&DS)**, Head of the *Department for Computer Science Engineering/ Department for Computer Science and Information Technology/Department of Artificial Intelligence and Data Science* for providing us with adequate facilities, ways and means by which we are able to complete this project-based Lab.

We thank all the members of teaching and non-teaching staff members, and also who have assisted me directly or indirectly for successful completion of this project. Finally, I sincerely thank my parents, friends and classmates for their kind help and cooperation during my work.

Hrushika Habeeb – 2520090054

Chakrika Panakala – 2520090042

TABLE OF CONTENTS

S.No.	Contents	Page no.
1.	Abstract	6
2.	Introduction	7
3.	Literature survey	8
4.	Probl3em Statement and Objectives	9
5.	Hardware and Software requirements	10
6.	Implementation	11
7.	Results & Discussion	12
8.	Conclusion & Future Scope	
9.	References	14

1. **ABSTRACT**

E-commerce platforms constantly face the challenge of identifying when listings from different sellers, marketplaces, or catalogs refer to the same underlying product despite inconsistent titles, attributes, images, and missing identifiers. This platform addresses the problem through a hybrid entity-resolution pipeline combining rule-based matching using SKUs, UPCs/GTINs, and brand-model patterns with fuzzy string matching, semantic embeddings, and image-based visual similarity.

A blocking and candidate-generation stage narrows the comparison space before pairwise or clustering-based matching, allowing the system to scale efficiently to catalogs containing millions of SKUs. Outputs include unified product IDs, deduplicated catalogs, and confidence-scored match links for price comparison, competitive intelligence, and marketplace onboarding.

The architecture integrates data ingestion, feature extraction, a scoring engine, and human-in-the-loop review for low-confidence matches. Key considerations include multilingual listings, variant differentiation, and balancing precision against recall.

Team Name: E-Commerce Product Entity-Resolution & Matching Platform

Team Members:

1. Hrushika Habeeb – 2520090054
2. Chakrika Panakala – 2520090042

2. INTRODUCTION

E-commerce marketplaces contain product listings created by many sellers and data sources. The same physical product may therefore appear under different titles, descriptions, attribute formats, identifiers, and images. A reliable method is needed to determine whether two records describe the same underlying product before catalogs are compared or consolidated.

The proposed E-Commerce Product Entity-Resolution & Matching Platform treats product matching as a multi-stage decision problem. It first ingests and normalizes listing data, then generates a manageable set of candidate pairs, extracts complementary textual, identifier, semantic, and visual features, and finally combines those signals into a match score. High-confidence matches can be unified automatically, while uncertain cases are routed to human review.

The project is motivated by practical applications such as price comparison, competitive intelligence, marketplace onboarding, catalog deduplication, and product information management. The remainder of this report presents related work, the problem and objectives, requirements, the proposed implementation modules, expected results, conclusions, future scope, and references.

3. LITERATURE SURVEY

1. DeepMatcher – Deep Learning for Entity Matching

DeepMatcher is a deep-learning framework for structured and textual entity matching. It provides data processing, neural matching, training, evaluation, and prediction workflows. It demonstrates how learned representations can complement traditional matching.

Reference: https://github.com/anhaidgroup/deepmatcher

2. Ditto – Deep Entity Matching with Pre-Trained Language Models

Ditto uses pre-trained Transformer language models and treats entity matching as sequence-pair classification. Its results show the value of contextual language representations when simple string similarity is insufficient.

Reference: https://arxiv.org/abs/2004.00584

3. Dedupe – Entity Resolution and Deduplication

Dedupe focuses on deciding whether records refer to the same real-world entity. Its documentation discusses normalization, strong identifiers, probabilistic matching, blocking, and threshold selection, all relevant to scalable catalog resolution.

Reference: https://dedupe.dev/guides/entity-resolution

4. Magellan – Entity Matching and Blocking

Magellan provides a workflow for candidate generation, feature extraction, and entity matching. Its blocking-oriented approach reduces unnecessary comparisons, which is important when catalogs contain large numbers of product records.

Reference: https://anhaidgroup.github.io/magellan/

5. Deep Learning for Entity Matching: A Design Space Exploration

Mudgal et al. study deep-learning approaches for structured, textual, and dirty entity-matching problems. The work motivates choosing matching methods according to data characteristics and combining learned features with established approaches.

Reference: https://pages.cs.wisc.edu/~anhai/papers1/deepmatcher-tr.pdf

4. Problem Statement and Objectives

Problem Statement: A clear and specific articulation of the problem, outlining its importance.

To design a scalable hybrid platform that identifies and links product listings representing the same underlying e-commerce product.

Problem Description: A concise explanation of the issue or challenge that needs to be solved.

Listings from different sources use inconsistent names, attributes, identifiers, and images, making direct equality unreliable and exhaustive comparison expensive.

Objectives:

Identify the Project's Purpose:

Build a unified workflow that ingests heterogeneous listings, generates candidates efficiently, combines similarity signals, assigns confidence scores, and supports human review.

Normalize titles, identifiers, brands, models, and attributes.

Use reliable SKU and UPC/GTIN identifiers where available. Apply fuzzy matching for spelling, punctuation, ordering, and formatting differences. Use semantic and image similarity as additional matching evidence.Use blocking to reduce expensive pairwise comparisons. Produce unified product IDs, deduplicated catalogs, and confidence-scored links. Support multilingual listings, product variants, and human review of uncertain cases.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

- Computer/laptop with a modern multi-core processor.
- At least 8 GB RAM recommended for development and catalog processing.
- Sufficient storage for product catalogs, features, model files, and product images.

Software Requirements:

- Operating System: Windows, Linux, or macOS.
- Programming environment for data ingestion, matching, and scoring.
- Libraries for tabular data, text normalization, fuzzy matching, feature extraction, and embeddings.
- Image-processing / vision library and a database or file-based catalog store.

Web or desktop interface for human review; version-control/development tools.

6. IMPLEMENTATION

```
MODULE 1 – DATA NORMALIZATION
def normalize(text):
  text = text.lower().strip()
  text = remove_punctuation(text)
  return collapse_spaces(text)

MODULE 2 – CANDIDATE GENERATION / BLOCKING
def candidates(record, index):
  key = make_blocking_key(record)
  return index.get(key, [])

MODULE 3 – HYBRID MATCH SCORING
def match_score(a, b):
  rule = identifier_score(a, b)
  fuzzy = fuzzy_text_score(a, b)
  semantic = embedding_similarity(a, b)
  visual = image_similarity(a, b)
  return combine(rule, fuzzy, semantic, visual)

MODULE 4 – DECISION AND REVIEW
score = match_score(product_a, product_b)
if score >= HIGH_THRESHOLD:
  create_unified_product_id(product_a, product_b)
elif score >= REVIEW_THRESHOLD:
  send_to_human_review(product_a, product_b, score)
else:
  mark_as_non_match(product_a, product_b)
```

7. RESULTS & DISCUSSION

The platform is designed to produce the following outputs from the matching pipeline:

1. Unified Product IDs – multiple source listings that represent the same product are linked to a common canonical identity.
2. Deduplicated Catalog – duplicate or overlapping listings can be grouped so that downstream systems operate on cleaner product data.
3. Confidence-Scored Match Links – every proposed link can carry a score that reflects the strength of the combined evidence.
4. Human Review Queue – low-confidence cases can be presented for manual confirmation instead of being merged automatically.

Discussion: The hybrid approach is intended to reduce the weaknesses of relying on one signal alone. Strong identifiers can provide high-precision matches, fuzzy matching handles small textual variations, semantic embeddings capture meaning, and visual similarity provides additional evidence when images are available. Blocking reduces the comparison space before expensive pairwise matching. The design therefore targets both matching quality and scalability, while explicitly considering false merges, missed matches, multilingual listings, and product variants.

8. CONCLUSION & FUTURE SCOPE

The E-Commerce Product Entity-Resolution & Matching Platform addresses the challenge of identifying duplicate or equivalent product listings across heterogeneous e-commerce sources. The proposed workflow combines deterministic identifiers, fuzzy text matching, semantic embeddings, and image similarity, with blocking used to improve scalability. The architecture also includes confidence scoring and human review so that uncertain matches are not forced into automatic decisions. These components directly support the project's goals of catalog deduplication, unified product identification, and reliable downstream product comparison.

Future Scope:

- Add stronger multilingual and cross-lingual semantic models for international catalogs.
- Improve product-variant handling so size, color, storage, generation, and bundle differences are not incorrectly merged.
- Introduce active learning so reviewer decisions continuously improve matching thresholds and models.
- Add distributed processing and scalable storage for very large catalogs containing millions of records.
- Improve image matching with stronger vision-language models and multiple product-image views.
- Provide analytics dashboards for match quality, precision, recall, review workload, and catalog coverage.
- Explore knowledge-graph representations to connect products, brands, models, categories, and sellers.

9. REFERENCES

1. AnHai Group, “DeepMatcher: Python package for performing Entity and Text Matching using Deep Learning.” https://github.com/anhaidgroup/deepmatcher

2. Y. Li, J. Li, Y. Suhara, A. Doan, W.-C. Tan, “Deep Entity Matching with Pre-Trained Language Models (Ditto).” https://arxiv.org/abs/2004.00584

3. Dedupe Documentation, “Entity Resolution, Explained Plainly.” https://dedupe.dev/guides/entity-resolution

4. AnHai Group, “Magellan: An Open-Source Toolkit for Entity Matching.” https://anhaidgroup.github.io/magellan/

5. S. Mudgal et al., “Deep Learning for Entity Matching: A Design Space Exploration.” https://pages.cs.wisc.edu/~anhai/papers1/deepmatcher-tr.pdf

6. “Extended ProMap datasets for product mapping.” Electronic Commerce Research. https://link.springer.com/article/10.1007/s10660-024-09892-9

7. Megagon Labs, “Ditto: Deep Entity Matching with Pre-Trained Language Models.” https://github.com/megagonlabs/ditto

8. Dedupe Documentation, “How it works.” https://docs.dedupe.io/en/stable/how-it-works/How-it-works.html

9. A. Steiner and C. Bizer, “Labeling Training Data for Entity Matching Using Large Language Models.” https://arxiv.org/abs/2606.28823

10. Q. Ruan, D. Shi, T. Bauernhansl, “Fine-tuning large language models with contrastive margin ranking loss for selective entity matching in product data integration.” https://doi.org/10.1016/j.aei.2025.103538